Fairness in Federated Learning:
Client-Side Interventions, Server Protocols & Adaptive Weighting

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1. Introduction

FedAvg can amplify demographic bias when client data are skewed. We present two linked studies:

- **189-run benchmark** of interventions (I_1–I_5) and server protocols (P_1–P_5).
- **DCA-FW**, which reduces the UTKFace EO gap by **40%** at baseline-level accuracy.

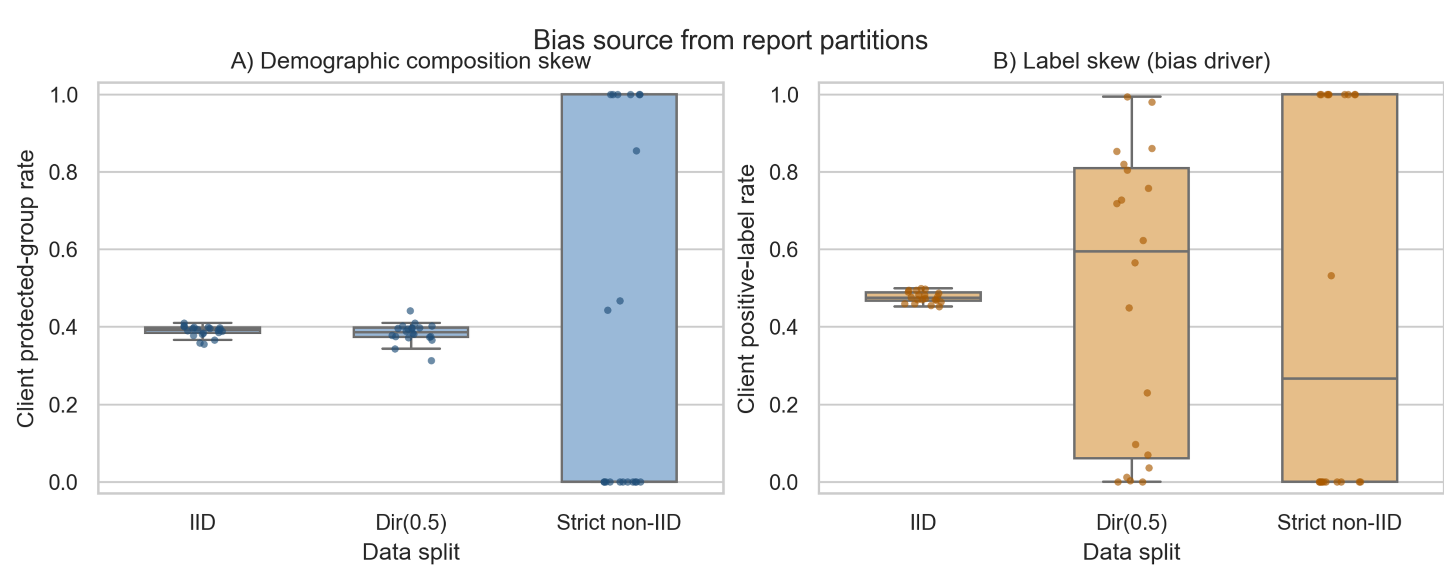
2. Problem Formulation

Setup: $K{=}20$ clients, local data $\mathcal{D}_k = \{(\mathbf{x}_i, y_i, s_i)\}_{i=1}^{n_k}$, with protected attribute `race_bin` $s_i \in \{0,1\}$. FedAvg objective:

$$\min_{\boldsymbol{\theta}} \sum_k \frac{n_k}{n} \mathcal{L}_k(\boldsymbol{\theta})$$

- **DP gap:** $|\mathbb{P}[\hat{y}{=}1 \mid s{=}0] - \mathbb{P}[\hat{y}{=}1 \mid s{=}1]|$
- **EO gap:** $\max_y |\mathbb{P}[\hat{y}{=}1 \mid y, s{=}0] - \mathbb{P}[\hat{y}{=}1 \mid y, s{=}1]|$
- Splits: **IID**, **Dirichlet-0.5**, **Strict non-IID**

3. Why Bias Emerges (from Report Data)



Heterogeneous splits induce demographic and label skew across clients.

4. Interventions and Protocols (Condensed)

Client interventions (I_1–I_5):

- **I_1:** loss reweighting
- **I_2:** gradient projection
- **I_3:** fairness-constrained objective
- **I_4:** representation disentanglement
- **I_5:** bi-level debiasing

Server protocols (P_1–P_5):

- **P_1:** fairness-audited aggregation
- **P_2:** gradient similarity filtering
- **P_3:** proximal fairness bounding
- **P_4:** representation fingerprinting
- **P_5:** stability testing

5. Novel: Dynamic Client-Adaptive Fairness Weighting

Designated fairness clients (30%) compute local race-wise accuracy $a_r^k(t)$, then adapt CE weights each round:

$$w_r^k(t{+}1) = [a_r^k(t) + \epsilon]^{-1} \quad \Longrightarrow \quad L_k = \textstyle\sum_r w_r^k \cdot L_{\mathrm{CE},r}^k$$

Outcome: client-level adaptation avoids FedAvg dilution and achieves strong fairness gains with near-zero accuracy trade-off.

Reference: Zhang, H., Liu, Y., He, X., Wu, J., Cong, T., & Huang, X. (2025). Sok: Benchmarking poisoning attacks and defenses in federated learning. arXiv preprint arXiv:2502.03801.

6. DCA-FW Experimental Setup

Parameter	Value
Datasets	UTKFace (Primary), FairFace (Validation)
Task	Binary gender classification (with race analysis)
Model Architecture	Custom Convolutional Neural Network
FL Framework	10 Clients, FedAvg Aggregation
Training Config	8–10 Comm. Rounds × 5 Local Epochs
Batch Size	32

7. Baseline: Optimizer × Split

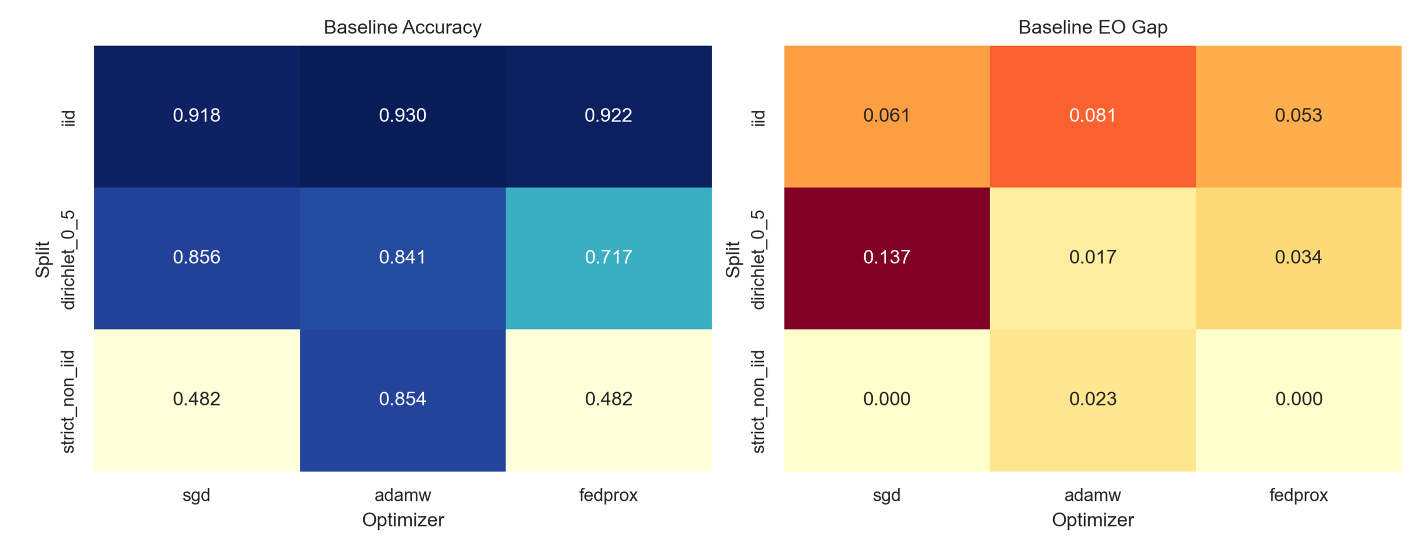


Figure 1. AdamW stable across splits (Acc ≥ 0.84); SGD/FedProx collapse on strict non-IID.

8. Intervention–Protocol Matrix (Acc / EO)

	I-only	P_1	P_2	P_3
I_1	.759/.028	.539/.059	.566/.060	.569/.073
I_2	.769/.043	.555/.059	.562/.070	.557/.063
I_3	.722/.034	.564/.066	.565/.071	.549/.069
	I-only	**P_3**	**P_4**	**P_5**
I_4	.628/.043	.567/.105	.590/.082	.561/.070
I_5	.683/.043	.568/.063	.562/.094	.535/.072

Table 1. No protocol dominates: P_4 → best acc (0.576) / worst EO (0.088); P_1 → best EO (0.061) / lower acc.

9. Key Finding: Optimizer Dominates Outcomes

Category	Optimizer	Acc	EO
I+P	AdamW	**0.773**	0.058
I+P	SGD	0.455	0.079
I+P	FedProx	0.454	0.078

Optimizer choice is the strongest factor. AdamW reaches 0.773 in I+P runs; SGD/FedProx collapse to ≈0.454 (driven by strict non-IID degenerate regimes).

10. Accuracy–Fairness Trade-off (189 Runs)

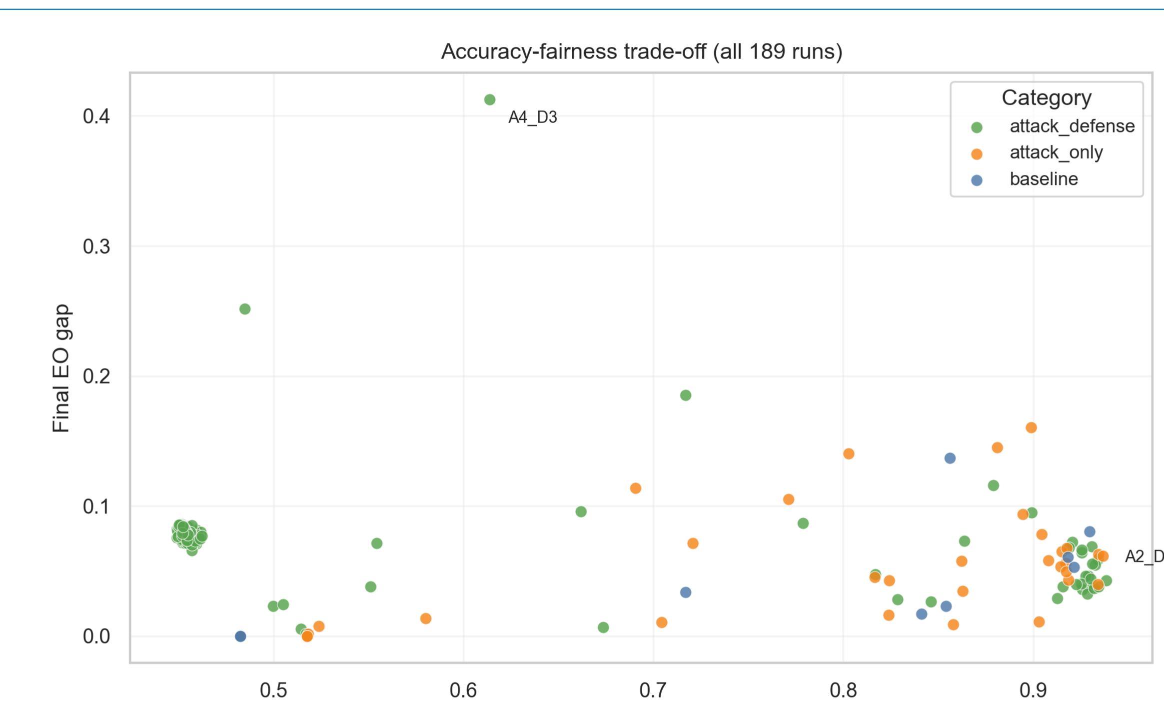


Figure 2. Best: I_2+P_3/IID/AdamW (Acc=.938, EO=.043). Worst EO: I_1+P_3/non-IID/AdamW (EO=.413).

11. DCA-FW Performance Comparison

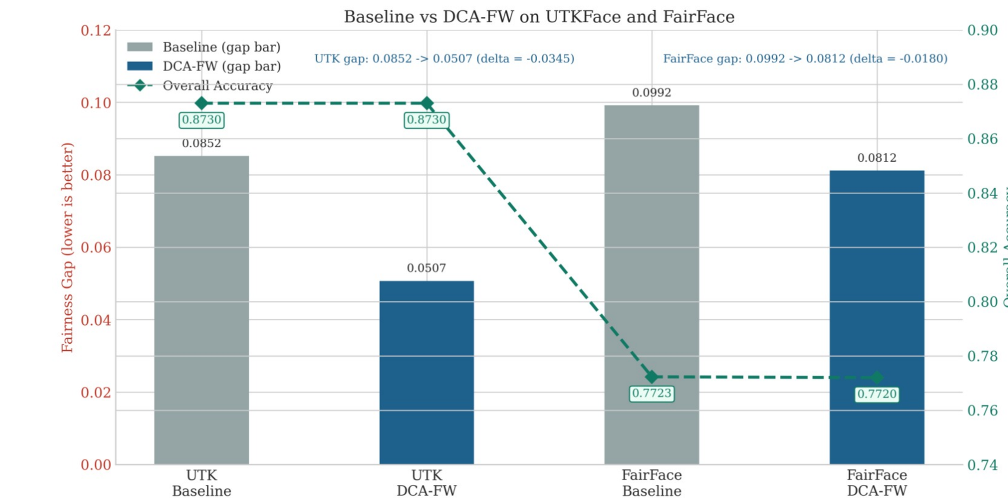


Figure 3. Fairness gap reduction under DCA-FW aggregation on UTKFace and FairFace datasets.

12. DCA-FW: Fairness Without Accuracy Loss

UTKFace — 5 Methods Compared

Method	Gap	Acc	ΔGap
Baseline FL	0.085	0.873	—
Static Weight	0.085	0.850	0%
Fairness Reg.	0.101	0.855	−19%
Adversarial	0.074	0.850	+14%
DCA-FW	**0.051**	**0.873**	**+40%**

FairFace Cross-Dataset Validation

Setting	Gap	Acc	ΔGap
Baseline	0.099	0.772	—
DCA-FW 40%	0.088	0.779	+11%
DCA-FW 50%	0.086	0.770	+14%
DCA-FW 60%	**0.081**	0.769	**+18%**

13. Unified Findings & Conclusions

1. **FL inherently amplifies demographic bias** even at baseline.
2. **Optimizer choice dominates** intervention–protocol outcomes; AdamW resists gradient distortions.
3. **No single protocol is universally optimal** — utility–fairness trade-offs need explicit policy.
4. **Representation interventions** (I_4, I_5) are most disruptive (ΔAcc ≤ −0.15) without proportional gains.
5. **DCA-FW** outperforms all methods by acting at the client level, avoiding FedAvg dilution.

⋆ MAIN FINDING ⋆

DCA-FW achieves a 40% fairness gap reduction on UTKFace without sacrificing baseline accuracy.

Future Work

Multi-seed · External baselines · Intersectional fairness · Convergence guarantees

14. Scan Full Reports

Report 1: Fairness Interventions & Server Auditing Protocols

Report 2: Dynamic Client-Adaptive Fairness Weighting

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